THANH P. LY

lyphucthanh0610@gmail.com • linkedin.com/in/phuc-thanh-ly/ • capfab.github.io
Google Scholar: profile • *ORCID:* 0009-0001-3420-5448

EDUCATION

Master of Science in Engineering (MSE) in Robotics 08/2023 – 05/2025
University of Pennsylvania, Philadelphia, USA

- Advisor: Rahul Mangharam
- GPA: 3.97/4.00

B.Eng. in Mechatronic Engineering, *summa cum laude* 08/2018 – 08/2022
Ho Chi Minh City University of Technology (HCMUT), Ho Chi Minh City, Vietnam — a member university of Vietnam National University – Ho Chi Minh City (VNU-HCM) and one of Vietnam's leading technical universities.

- Thesis: "Integration of an Industrial Robot with a 3D Scanning Stereo Vision System Using Gray Code and Phase Shift."
- Advisor: Quoc Chi Nguyen
- GPA: 8.87/10

RESEARCH & PROFESSIONAL EXPERIENCE

VinRobotics **Hanoi, Vietnam**
Humanoid robotics subsidiary of Vingroup, Vietnam's largest private conglomerate (VinFast, Vinhomes, Vinpearl).

Acting Head of Reinforcement Learning 04/2026 – Present

- Scope: lead a team of 12, including interns; report to the Director of Foundation AI. Own use-case development; contribute to the group's research, source code, benchmarks and evaluation protocols.
- Developing a motion planning system for whole-body-controlled humanoid teleoperation, fusing occupancy-map and whole-body state estimation into a point-to-point navigation planner for cluttered environments. Closing the loop with an RL-trained whole-body controller by injecting planner motion sequences as reference trajectories, yielding natural, dynamically consistent whole-body motion during traversal.
- Developed **TACT-ful**: a perceptive locomotion system for payload-robust humanoid walking, combining terrain-aware foothold planning with reward shaping for payload compliance. Achieves 1.0 m/s on stairs with 20 cm risers and zero-shot transfer to physical humanoid carrying up to 20 kg, validated over 1 hour of real-robot testing. Project page.
- Co-developed **CompliantWBC**: whole-body compliance control for a 70 kg heavy humanoid under high-payload and external-force scenarios, combining impedance control with learned force estimation. Validated on surface wiping, force reaction, loaded squatting, and cooperative transport with 0.98 success rate. Project page.
- Led development of a terrain-classifier-routed mixture-of-experts locomotion system with model-based foothold planning and gradient-based refinement, eliminating dependence on motion-capture retargeting and generative motion priors. Achieved 90% success on an 85 m, five-segment multi-terrain course in a deployment-fidelity simulator. Project page.

- Mentored 11 interns across locomotion policy design, motion imitation, sim-to-sim and sim-to-real transfer, reward engineering, and pipeline and asset validation. (Full breakdown in Teaching & Mentorship, below.)

Senior Reinforcement Learning Engineer 12/2025 – 03/2026

- Developed and cross-validated multi-physics simulation environments in Isaac Lab and mjlab, across the PhysX, MuJoCo, and Newton/Warp physics engines.
- Built high-throughput parallel simulation pipelines supporting multi-DOF humanoid embodiments for large-scale distributed RL training on H100 GPU clusters — 4,096 parallel environments for standard locomotion policies, 1,024 for perceptive (terrain-conditioned) policies — reaching a deployable policy in as little as 12 hours.
- Collaborated with hardware and controls teams to design policies compatible with low-level joint PD controllers, torque limits, and safety constraints, directly informing actuator delay modeling and dynamics randomization strategies.

R&D Software Engineer – Reinforcement Learning Engineer 06/2025 – 11/2025

- Designed and trained RL policies for humanoid locomotion across varied terrain and disturbance conditions, achieving stable walking, recovery, and adaptive balancing on physical hardware.
- Developed custom locomotion environments in Isaac Gym and Isaac Lab with event-driven gait initialization and phase-based curriculum learning to accelerate convergence and transfer.
- Implemented reward functions targeting stability, energy efficiency, natural motion quality, and smooth sim-to-real transfer for high-dimensional continuous control.
- Conducted systematic domain randomization experiments across dynamics, contact, and sensing parameters to characterize and reduce sim-to-real gap on real humanoid deployments.

University of Pennsylvania **Philadelphia, USA**

Research Assistant 05/2024 – 05/2025

- Generated large-scale synthetic visual datasets (1M+ photo-realistic images) in NVIDIA Omniverse, enabling robot perception models deployable in real warehouse environments with sub-30 cm localization accuracy.
- Improved multi-view pose estimation by 40% through combined synthetic and real-data fine-tuning.
- Developed homography-based multi-camera tracking and 3D reconstruction pipelines deployed in live environments; automated ground-truth labeling using Vicon motion capture for scalable training data collection.
- Calibrated multi-camera and structured-light stereo rigs for dense 3D reconstruction and depth estimation, and fused outputs across sensors for robot perception.

Teaching Assistant – Computer Vision & Computational Photography 08/2024 – 12/2024

- Led lab sessions and provided technical support for projects on multi-view geometry, camera calibration, and image-based reconstruction. (Course details in Teaching & Mentorship, below.)

Procter & Gamble **Ho Chi Minh City, Vietnam**

Process Engineer – Power Control & Information Systems 01/2023 – 05/2023

- Developed digital twins and applied machine learning models to optimize energy consumption in a large-scale industrial facility.

Vingroup Big Data Institute (VinBigData) **Hanoi, Vietnam**

AI and big-data research institute under Vingroup, Vietnam's largest private conglomerate.

Technology Specialist 07/2022 – 12/2022

- Researched monocular depth estimation methods for 3D reconstruction in robotic vision.
- Developed vision-based safety monitoring pipelines for construction environments.

Ho Chi Minh City University of Technology **Ho Chi Minh City, Vietnam**

Undergraduate Research Assistant 04/2021 – 06/2022

- Developed 3D reconstruction and calibration pipelines using active structured light and dual-quaternion formulations for precise robot–camera alignment.
- Automated image acquisition, stereo calibration, and grasp execution for a robotic manipulation pipeline.
- Applied PointNet-based deep learning for bin-picking part segmentation.

Intel **Ho Chi Minh City, Vietnam**

Undergraduate Technical Intern 02/2021 – 08/2021

- Supported equipment qualification and data analysis for chipset production.

PUBLICATIONS

Peer-Reviewed Conference & Journal Papers

1. **Thanh Ly** (first author), Quoc Chi Nguyen, Ngoc Duy Hung Nguyen, Phuong-Tung Pham, and Keum-Shik Hong, "Structured-Light-Based 3D Scanning System for Industrial Manipulator in Bin-Picking Application," *Australian & New Zealand Control Conference (ANZCC)*, 2022, pp. 34–39. DOI: 10.1109/ANZCC56036.2022.9966979.
2. Ngoc Duy Hung Nguyen, Le Hai Nguyen Nguyen, Phuong-Tung Pham, Quoc Chi Nguyen, and **Thanh Ly**, "Bin-Picking Solution for Industrial Robots Integrating a 2D Vision System," *International Conference on High Performance Big Data and Intelligent Systems (HDIS)*, 2022, pp. 266–270. DOI: 10.1109/HDIS56859.2022.9991341.

Workshop Papers

1. **Thanh Ly** (co-first author), Truong-Duy Dang, Chien Le, Tan-Dzung Do, Tuan Dat Phuong, Cuc T. Trinh, Vien Anh Ngo, and An T. Le, "TACT-ful: Multi-Channel Terrain Affordance and Compliance Training for Payload-Robust Perceptive Humanoid Locomotion." Accepted at the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2026 Workshop on Full-Shift Robot Co-Workers. arXiv:2606.20645 (06/2026). Project page.

Manuscripts Under Review

1. **Thanh Ly** (co-first author), Chien Le, Truong-Duy Dang, Duy M. H. Nguyen, Duy-Dung Le, Vien Anh Ngo, and An T. Le, "Terrain Traversal Without Motion Priors: Expert-Routed Humanoid Locomotion at Deployment Fidelity." Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. Submitted 09/2026. arXiv:2610.14061 (10/2026). Project page.

2. Quynh Anh Huynh, **Thanh Ly**, Hoang M. Truong, Cuc T. Trinh, Chien Le, Georgia Chalvatzaki, and An T. Le, "TD-MTP: Coupling Structured Planning with Multi-Step Value Learning." Submitted to the *International Conference on Learning Representations (ICLR)*, 2027. Project page.
3. Tai Hoang, Huy Le, **Thanh Ly**, Truong-Duy Dang, Chien Le, Aleksandar Taranovic, Gerhard Neumann, An T. Le, and Vien Anh Ngo, "Pretraining Foundation Policies for Perceptive Humanoid Locomotion with Offline RL." Submitted to the *International Conference on Learning Representations (ICLR)*, 2027. Project page.
4. Tan-Dzung Do, Cuc T. Trinh, Tuan Dat Phuong, Chien Le, **Thanh Ly**, Vien Anh Ngo, and An T. Le, "CompliantWBC: Whole-Body Compliance for Heavy Humanoids via Force Latent Estimation and Residual Impedance Targets." Submitted to *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. Submitted 09/2026. arXiv:2609.33310 (09/2026). Project page.

Preprints

1. TACT-ful (above, Workshop Papers): arXiv:2606.20645, 06/2026.
2. Terrain Traversal Without Motion Priors (above, Manuscripts Under Review): arXiv:2610.14061, 10/2026.
3. CompliantWBC (above, Manuscripts Under Review): arXiv:2609.33310, 09/2026.

Technical Reports & Course Projects
(Course deliverables, not peer-reviewed – kept visually separate from the sections above.)

1. P. A. Hsieh, **Thanh Ly**, X. H. Ong, "Reinforcement Learning for Autonomous Drone Racing," Final Project Report, Learning in Robotics, University of Pennsylvania, 2024.
2. P. Do, **Thanh Ly**, X. H. Ong, "Mini Minecraft," Final Project, Interactive Computer Graphics, University of Pennsylvania, 2024. [Online]. Available: https://youtu.be/Qss9ZLcg4YA
3. **Thanh Ly**, X. H. Ong, Z. Osman, "Stable Diffusion 3 Visualization Using NCUT," Final Project Report, Advanced Topics in Machine Perception, University of Pennsylvania, 2024.

Theses

1. **Thanh Ly**, "Integration of an Industrial Robot with a 3D Scanning Stereo Vision System Using Gray Code and Phase Shift," B.Eng. Thesis, Ho Chi Minh City University of Technology, 2022. Advisor: Quoc Chi Nguyen.

AWARDS & HONORS

- **Excellent Thesis Award**, Ho Chi Minh City University of Technology (HCMUT), 09/2022.

SCHOLARSHIPS & FELLOWSHIPS

- **Vingroup Science and Technology Scholarship Program for Overseas Study**, awarded 2022, supporting MSE studies at the University of Pennsylvania, 2023 – 2025. One of 30 recipients from about 1,000 applicants.
- **Panasonic Scholarship**, 2022. Cohort of 20 recipients.
- **Saigon Hi-Tech Park Business Association (SBA) Scholarship**, 2022. SBA is the association of companies in Saigon Hi-Tech Park, a state-run technology zone in Ho Chi Minh City, Vietnam.
- **KSYS-CUBE Scholarship**, 2022, funded jointly by Kanden Systems Co., Ltd. and Cube System Inc. (Japan).

PROFESSIONAL SERVICE

- Reviewer, IEEE International Conference on Robotics and Automation (ICRA), 2027.

TEACHING & MENTORSHIP

Teaching

- Teaching Assistant, *Computer Vision & Computational Photography* (CIS 5810), University of Pennsylvania. Level: graduate. Term: Fall 2024 (08/2024 – 12/2024). Led lab sessions and provided technical support for projects on multi-view geometry, camera calibration, and image-based reconstruction.

Mentorship

- VinRobotics: mentored 11 interns, 04/2026 – 08/2026. Topics: locomotion policy design, motion imitation, sim-to-sim and sim-to-real transfer, reward engineering, and pipeline and asset validation.

INVITED TALKS & PRESENTATIONS

- Poster, TACT-ful, IEEE/RSJ IROS 2026 Workshop on Full-Shift Robot Co-Workers, Pittsburgh, USA, 09/2026.

MEDIA & PRESS

- Featured in "From Vietnam to Penn: Thanh Ly's Journey as an International Student in Robotics," Penn Engineering Graduate Admissions, University of Pennsylvania, January 9, 2025. Article.

TECHNICAL PROJECTS

Reinforcement Learning for Autonomous Drone Racing
Learning in Robotics, University of Pennsylvania

- Designed and trained RL policies in custom simulation environments for agile drone navigation under dynamic constraints; investigated sim-to-real policy robustness.

RetroLifts – Infrastructure-Based Autonomous Forklift System

- Designed infrastructure-mounted perception architecture combining BEV global localization with FPV local planning, reducing onboard sensing and compute requirements.
- Implemented multi-camera calibration and multi-view geometry pipelines for accurate global object localization in warehouse environments.
- Built simulation models for motion planning and collision checking to support scalable deployment analysis.

Mini Minecraft
Interactive Computer Graphics, University of Pennsylvania

- Built a real-time 3D voxel rendering engine in C++ with a custom OpenGL pipeline, including chunk-based procedural terrain generation.
- Implemented GLSL vertex and fragment shaders for dynamic lighting, shadows, and physically-based material rendering.

TECHNICAL SKILLS

Sim-to-Real	Domain randomization, dynamics & actuator delay sim, contact modeling, symmetry augmentation, real-robot validation
Simulation Frameworks	Isaac Lab, Isaac Sim, Isaac Gym, mjlab
Physics Engines	PhysX, MuJoCo, Newton/Warp
RL & Control	PPO / policy optimization, reward engineering, whole-body locomotion, terrain-aware foothold planning, disturbance recovery, contact-rich control
Perception	Multi-view geometry, camera calibration, sensor fusion, synthetic data generation, 3D reconstruction, depth estimation
Software	Python, C++, PyTorch, Slurm, distributed GPU training
Sensing Systems	Contact sensors, ray-cast terrain, height maps, depth cameras, Vicon/Qualisys/Nokov motion capture
Rendering & Graphics	OpenGL, GLSL shaders (vertex/fragment), procedural geometry pipelines, lighting & shadow systems, texture/material rendering

LANGUAGES

Vietnamese, English, German (elementary).

REFERENCES

Available upon request.